

SAURAV MALLA

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PROFESSIONAL SUMMARY

Results-driven analytical chemist with 5+ years of experience in analytical testing, method development, and research. Expertise in operating advanced instrumentation including ICP-MS, HPLC, and spectroscopy equipment. Strong background in R&D projects with proven ability to optimize processes and solve complex analytical challenges. Self-motivated learner with intermediate R programming and basic SQL skills.

TECHNICAL SKILLS

Analytical Techniques: ICP-MS, HPLC, GC/MS, Atomic Absorption Spectroscopy (GFAA), Fluorescent Spectroscopy, NMR, Ion Chromatography (Cation/Anion IC)

Characterization Methods: Dynamic Light Scattering (DLS), Differential Scanning Calorimetry (DSC), Fluorometry

Microscopy: Leica Stereomicroscope, Olympus Fluoview Microscope

Programming & Software: R (Intermediate), SQL (Basic), Data Analysis

Laboratory Skills: Method Development, QC Testing, Reagent Preparation, Troubleshooting, Microfluidic Device Fabrication

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

University of North Dakota, Department of Chemistry | Grand Forks, ND | August 2020 – March 2021

- Conducted advanced research regarding occurrence of both nonvolatile and semi volatile carbonaceous air particulate markers using thermal desorption-pyrolysis-gas chromatography-mass spectrometry.
- Collaborated with research team on experimental design and data analysis.

Analytical Chemist

Kelly Scientific Inc. (SITE: SA CHEM INC.) | Cleburne, TX | June 2019 – February 2020

- Conducted analytical testing of products using ICP-MS, HPLC, atomic absorption spectroscopy, and ion chromatography.
- Performed routine instrument troubleshooting and maintenance to ensure optimal performance and data quality.
- Prepared reagents and maintained laboratory standards in compliance with safety protocols.
- Contributed to R&D projects focused on analytical method development and optimization.
- Documented and reported results with accuracy and attention to detail.

Graduate Researcher

University of Oklahoma | Norman, OK | August 2015 – July 2018

- Developed and optimized ruthenium thin films using spin coating techniques with full characterization.
- Designed and fabricated PDMS microfluidic devices for analytical applications.

R&D Chemistry Intern

Senseonics Inc. | Germantown, MD | June 2014 – September 2014

- Executed QC test runs on post-clinical glucose sensors across various concentration ranges.
- Conducted fluorescence measurements and qualitative protein analysis using fluorometry.
- Performed boronate group analysis in indicator gels for continuous glucose monitoring systems.

Undergraduate Researcher

Southeastern Louisiana University | Hammond, LA | July 2010 – August 2012

- Synthesized dihydroquinolines via modified Skraup cyclization.
- Characterized liquid and solid products from wood gasifier using DLS.
- Performed thermal analysis of DMSO using differential scanning calorimetry.
- Conducted fractionation studies of EPDM rubber using centrifuge techniques.

EDUCATION

Master of Science in Chemistry (Analytical Concentration)

University of Oklahoma | Norman, OK | Graduated Fall 2018

GPA: 3.76/4.0

Bachelor of Science in Chemistry, Minor in Biology

Southeastern Louisiana University | Hammond, LA | Graduated Fall 2013

GPA: 3.42/4.0

PRESENTATIONS & CONFERENCES

- **SWRM/SERMACS 2010 Conference** – Poster Presentation, New Orleans, LA (December 2010)
- **Southwest Regional Meeting (SWRM)** – Poster Presentation, Austin, TX (November 2011)
- **Chemistry Conference, University of Louisiana** – Poster Presentation, Lafayette, LA (November 2010)

AWARDS & RECOGNITION

Provost's Certificate of Distinction in Teaching | University of Oklahoma | 2016, 2017

LEADERSHIP & ENGAGEMENT

Vice President, Nepalese Student Association, University of Oklahoma (August 2016 – August 2018)

Graduate Teaching Assistant, University of Oklahoma (August 2015 – August 2018)

Supplemental Instruction Leader, Southeastern Louisiana University (January 2013 – December 2013)

General Member, Southeastern Chemistry Club (January 2011 – December 2013)

COMMUNITY INVOLVEMENT

- **You Be the Chemist**, Southeastern Louisiana University (2011, 2012)
- **Super Science Saturday**, Louisiana State University (October 2010)